Walchand College of Engineering, Sangli
Department of Computer Science and Engineering

Practical No. 6

Study and implementation of node.js

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Batch - S5

Problem Statement 1: Introduction to Node.js

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1. What is Node.js? Difference from Apache/PHP?

Answer:

- Node.js is a **JavaScript runtime environment** built on Chrome's V8 engine.
- It allows running JavaScript **outside browser (server-side)**.

Apache/PHP:

- Apache is a **web server**
- PHP is a **server-side language**

Key Difference:

- Node.js uses **event-driven, non-blocking I/O**
- Apache/PHP use **blocking, thread-based model**

Meaning:

- Node.js handles **many users with single thread**
- PHP creates **new thread per request**

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2. Purpose of V8 Engine

Answer:

- V8 engine converts **JavaScript → machine code**
- Written in C++
- Makes execution **very fast**

That's why Node.js is fast compared to traditional systems

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3. Single-threaded Event-driven Architecture

Answer:

- Node.js runs on **single thread**
- Uses **event loop** to manage tasks

Flow:

Request → Event Loop → Callback Queue → Execution

It doesn't create multiple threads
Instead, it handles multiple requests using **callbacks/events**

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4. Why Node.js is Non-blocking?

Answer:

- Node.js does not wait for tasks (like file read, DB call)
- It continues execution

Example:

```
fs.readFile('file.txt', ()=>{});
console.log("Next line executes immediately");
```

File loads in background → program continues

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5. What is npm?

Answer:

- npm = **Node Package Manager**
- Used to install/manage libraries

Uses:

- Install packages
- Manage dependencies
- Run scripts

```
npm install express
```

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6. What is Module?

Answer:

- A module is a **separate file containing reusable code**

Export:

```
module.exports = add;
```

Import:

```
const add = require('./file');
```

Helps in **code organization**

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7. require() vs import

require()	import
CommonJS	ES Modules
Default in Node	Modern JS
Synchronous	Asynchronous

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8. Custom Module

Example:

```
// math.js
function add(a,b){
return a+b;
}
module.exports = add;
```

Then use:

```
const add = require('./math');
```

---

9. package.json

Answer:

- It is the **configuration file of Node project**

Contains:

- Project name
- Dependencies
- Scripts

Created using:

```
npm init -y
```

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10. Install packages

Local:

```
npm install express
```

Used only in project

Global:

```
npm install -g nodemon
```

Used anywhere in system

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11. Async vs Sync

Async	Sync
Non-blocking	Blocking
Faster	Slower
Uses callback	Sequential

Node prefers **async**

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12. Create HTTP Server

```
const http = require('http');
http.createServer((req,res)=>{
res.write("Hello");
res.end();
}).listen(3000);
```

Basic server without Express

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13. http vs Express

http	Express
Low-level	High-level
Manual work	Easy
Complex	Simple

Express = shortcut over http

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14. Handle GET & POST

```
if(req.method === "GET"){
res.end("GET request");
}
if(req.method === "POST"){
res.end("POST request");
}
```

In Express → easier

Problem Statement 2: Middleware

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1. What is Middleware?

Answer:

- Function that runs **between request and response**

Used for:

- Logging
- Authentication
- Validation

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2. Custom Middleware

```
const logger = (req,res,next)=>{
console.log(req.url);
next();
};
app.use(logger);
```

`next()` is compulsory

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3. Execution Order

Answer:

- Middleware runs in **sequence**
- Order matters

Example:

```
app.use(A);
app.use(B);
```

A runs first, then B

Problem Statement 3: File System (fs)

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1. Read & Write File

```
fs.readFile('file.txt','utf8',(err,data)=>{
console.log(data);
});

fs.writeFile('file.txt','Hello',()=>{});
```

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2. readFile vs readFileSync

- readFile → async (non-blocking)
- readFileSync → sync (blocking)

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3. Check File Exists

```
fs.existsSync('file.txt');
```

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4. Async File Handling

- Uses callbacks/promises
- Does not block execution

Problem Statement 4: Database

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1. Connect MySQL

```
const mysql = require('mysql2');
const db = mysql.createConnection({
host:'localhost',
user:'root',
password:'',
database:'test'
});
```

---

2. mysql2

- Library to connect Node + MySQL
- Supports promises

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3. CRUD

Create:

```
INSERT INTO users VALUES(1,'Atharv');
```

Read:

```
SELECT * FROM users;
```

Update:

```
UPDATE users SET name='A';
```

Delete:

```
DELETE FROM users;
```

Problem Statement 5: REST API

API Routes

POST /books
GET /books
PUT /books/:id
DELETE /books/:id

Features

- Add book
- Update book
- Delete book
- Filter data

Authentication

- Use middleware
- JWT / session

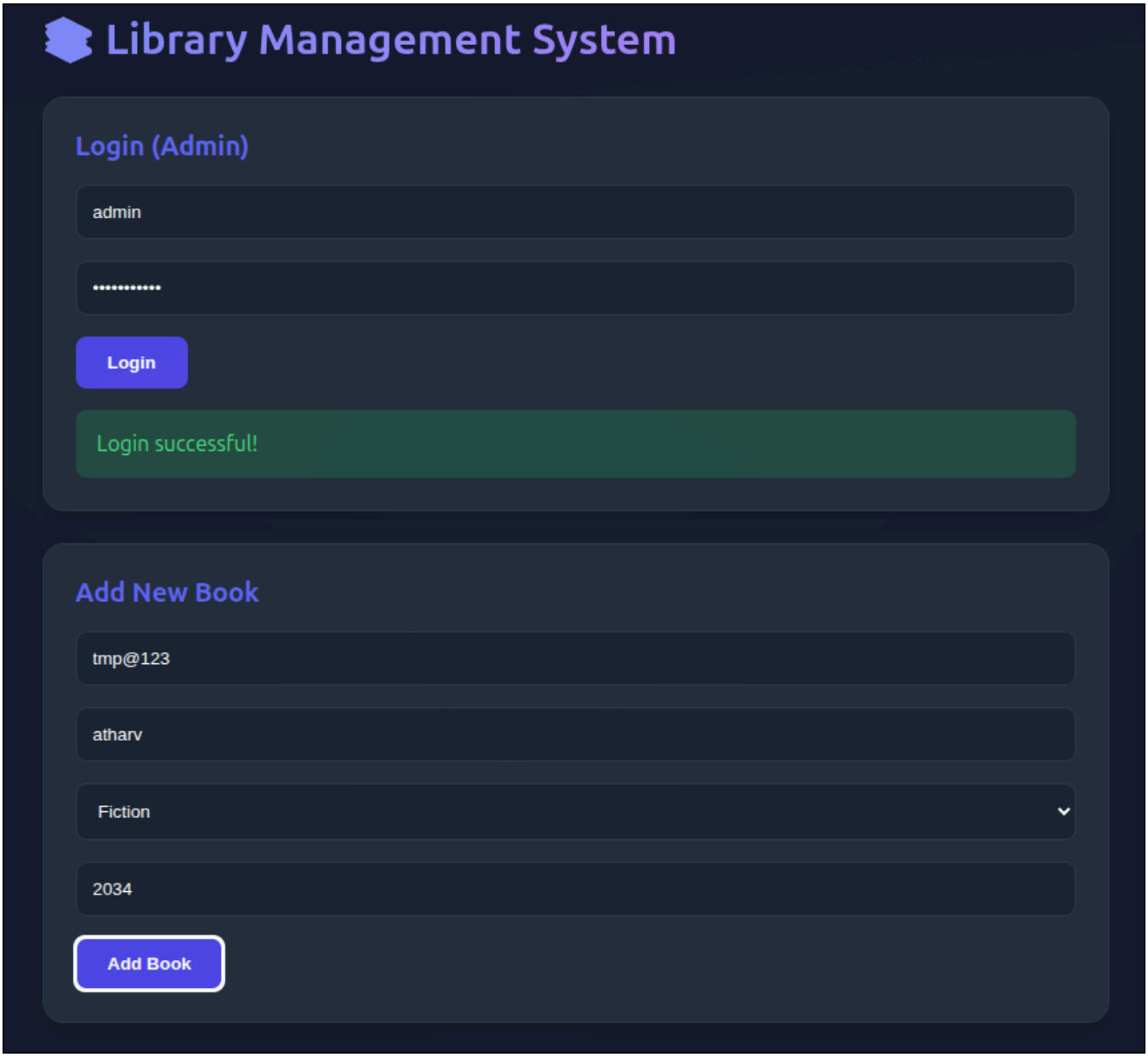
File Upload (Multer)

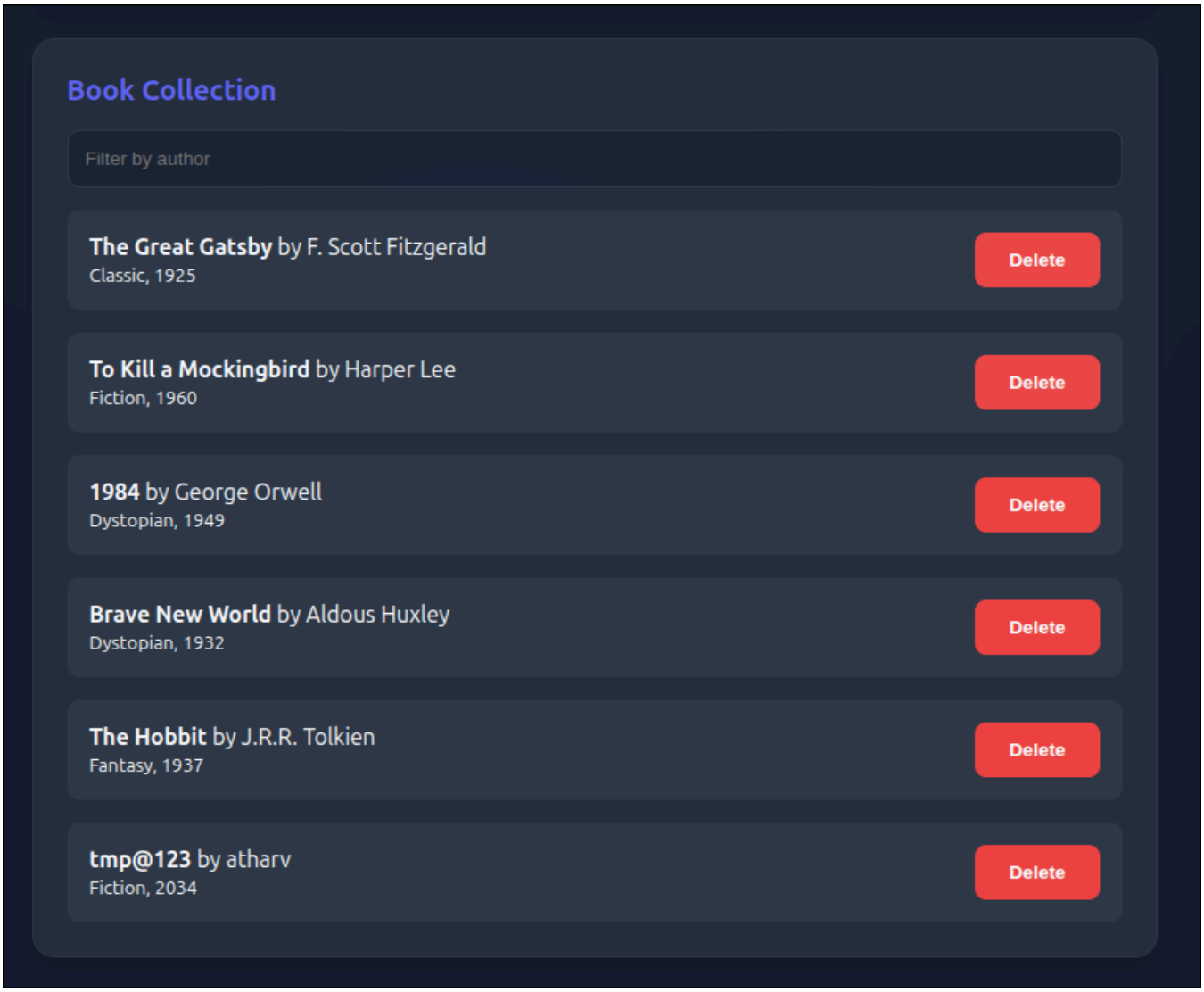
```
const multer = require('multer');
const upload = multer({ dest:'uploads/' });

app.post('/upload', upload.single('file'), (req,res)=>{
res.send("Uploaded");
});
```

Features

- Upload
- View
- Download/Delete
- Auth protected

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